Euclid's Elements — Book I
Proposition 24

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If two triangles have two sides equal to two sides respectively, but have one of the angles contained by the equal straight lines greater than the other, they will also have the base greater than the base.

1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the formal development, the hypotheses are

$$AB \cong DE, \qquad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced. Proposition I.24 is nevertheless substantially more delicate than its statement suggests. The classical diagram suppresses a genuine case distinction in the position of the auxiliary point. The Lean reconstruction makes this order-theoretic issue explicit and proves all possible configurations.

This is also a historically significant feature of I.24. The proof printed in the traditional text of Euclid treats only one of three possible configurations. Proclus supplied arguments for the omitted configurations, and later editors and commentators discussed alternative repairs. Thus the case split forced by the formal reconstruction is not an artifact of Lean: it reflects a classical issue in the textual and mathematical history of the proposition.

2 The Classical Configuration

Let ABC and DEF be nondegenerate triangles satisfying

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle EDF < \angle BAC.$$

Euclid constructs at D an angle congruent to the larger angle at A and lays off a segment DG congruent to AC (and hence to DF). Thus

$$\angle EDG \cong \angle BAC, \qquad DG \cong AC \cong DF.$$

By SAS,

$$\triangle ABC \cong \triangle DEG,$$

so that

$$BC \cong EG.$$

It remains to prove

$$EF < EG.$$

The historical difficulty is best seen by displaying all three configurations distinguished in modern commentary. Following Wilkins, the distinction is made by the positions of D and E relative to the line through G and F.

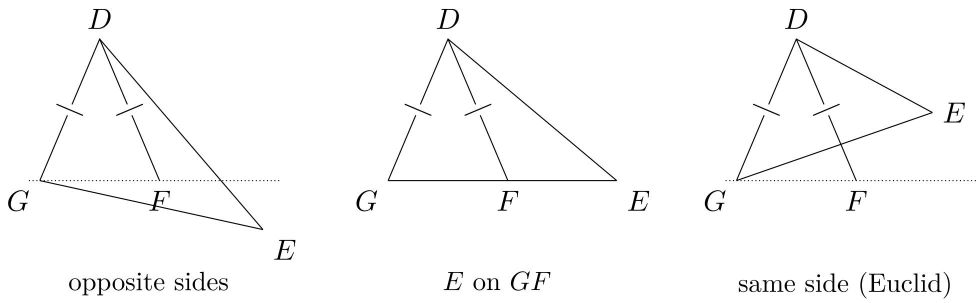

Figure 1: The three configurations distinguished in the classical discussion of Proposition I.24. Euclid's printed proof treats only the third.

The equal marks on DG and DF record the isosceles triangle used in the classical argument. The three diagrams are not three Lean branches; they belong to the historical analysis of Euclid's construction.

3 Classical Proof

Euclid's argument proceeds as follows. Since

$$AB \cong DE, \qquad AC \cong DG, \qquad \angle BAC \cong \angle EDG,$$

Proposition I.4 gives

$$BC \cong EG.$$

Also

$$DF \cong DG,$$

so triangle DFG is isosceles. Proposition I.5 therefore supplies the corresponding equality of its base angles. In the configuration displayed by Euclid, this angle equality, together with the position of E, yields a strict comparison between the two angles of triangle EFG. Proposition I.19 then gives

$$EF < EG.$$

Since $EG \cong BC$, one obtains

$$EF < BC.$$

The classical dependency pattern is therefore

$$\begin{array}{c}
\angle EDF < \angle BAC \\
\downarrow \\
\text{construct } G: \quad DG \cong AC, \quad \angle EDG \cong \angle BAC \\
\downarrow \ I.4 \\
EG \cong BC \\
\downarrow \\
DF \cong DG \\
\downarrow \ I.5 \\
\text{angle comparison in } \triangle EFG \\
\downarrow \ I.19 \\
EF < EG \\
\downarrow \\
EF < BC.
\end{array}$$

The problem is not the metric or congruence part of this argument. The problem is the unproved assumption about the relative position of the auxiliary points.

4 The Classical Configuration Gap

Modern commentary makes the difficulty precise. After the construction used in I.24, three configurations are possible. In Wilkins's formulation they are distinguished by the position of the relevant points with respect to the line through the two auxiliary points. The proof given by Euclid covers only one of these configurations; Proclus supplied proofs for the other two.

This point was also discussed by later editors. Simson modified the setup by adding a restriction on which of the two equal sides is chosen in the construction, with the aim of avoiding the three-case analysis. Heath's commentary records both the added condition and the underlying issue.

The historical point should therefore be stated carefully. Hilbert did not first discover the missing cases of I.24. They were already known in the ancient commentary tradition. What the Hilbert axiomatic framework provides is a language in which the relevant incidence, separation, and betweenness information must be stated and proved rather than inferred from a picture.

The Lean reconstruction follows this axiomatic discipline. It does not assume that the auxiliary point occupies the position suggested by one particular diagram.

For the third configuration, which is the one explicitly treated by Euclid, Wilkins makes the angle comparison visually explicit. Since

$$DG \cong DF,$$

the base angles of the isosceles triangle DGF are congruent, and the displayed configuration gives the chain

$$\angle EFG > \angle DFG = \angle DGF > \angle EGF.$$

Proposition I.19 then yields $EG > EF$.

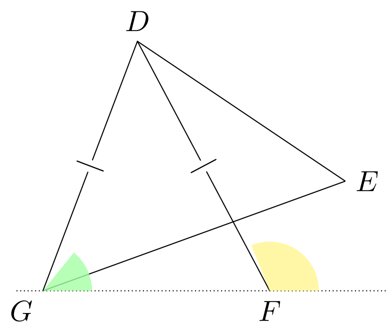

Figure 2: Euclid's third configuration, with the two angle-comparison regions highlighted schematically. The colours are explanatory only; the formal proof uses synthetic angle order, not numerical angle measure.

5 Strict Angle and Segment Comparison

Two synthetic order relations are central to the reconstruction.

The relation `HilbertAngleLess` expresses strict comparison of angles without introducing numerical angle measure. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

Similarly, `HilbertSegmentLess` expresses strict comparison of segments through betweenness and congruence, without introducing a real-valued length function.

Proposition I.24 combines these two order theories. Its hypothesis is a strict angle inequality, while its conclusion is a strict segment inequality. The hinge construction is the bridge between them.

6 Proof Architecture of the Reconstruction

The formal proof separates naturally into seven stages.

6.1 The SAS Construction

The helper `euclid_proposition_24_sas_aux` starts from

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle EDF < \angle BAC.$$

The strict angle comparison supplies an interior ray of $\angle BAC$. A point H is laid off on this ray so that

$$AH \cong AC.$$

Since $AC \cong DF$, congruence transport gives

$$AH \cong DF.$$

The auxiliary angle at A is congruent to $\angle EDF$, and SAS applied to ABH and DEF yields

$$BH \cong EF.$$

Thus the original goal $EF < BC$ is reduced to the hinge comparison

$$BH < BC.$$

6.2 The Hinge Configuration

The interior ray through H meets the open segment BC at a point Y. The same-ray information does not determine the order of H and Y on the ray from A. The formal proof therefore uses same-ray trichotomy and obtains exactly three possibilities:

$$A - H - Y, \qquad H = Y, \qquad A - Y - H.$$

This is the central order-theoretic decomposition of the reconstruction.

It is important not to identify these three Lean cases with the three historical configurations above. The historical split concerns the position of D and E relative to the line GF. The Lean split concerns the order of H and Y on a single ray from A after the auxiliary hinge point has already been constructed.

The formal geometry can be displayed independently of the historical diagram:

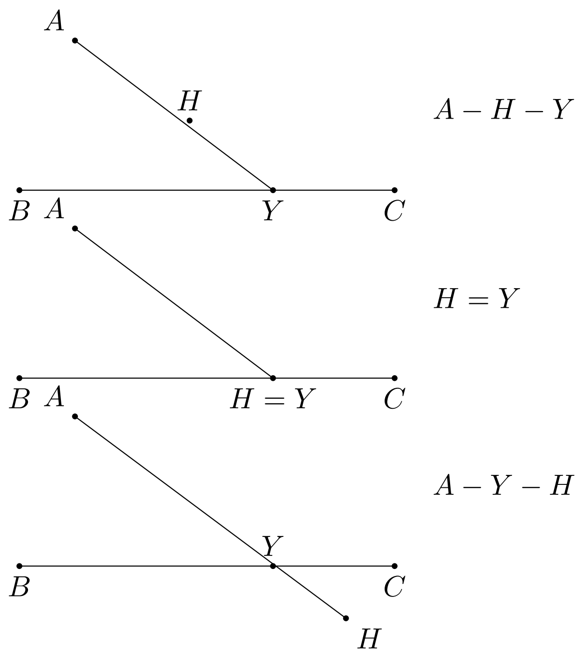

Figure 3: The three order configurations used by the Lean hinge helper. In all three, Y lies in the open segment BC; the split concerns only the order of H and Y on their common ray from A.

6.3 Case 1: $A - H - Y$

This is the longer nontrivial branch. After reconstructing the necessary noncollinearity facts, Proposition I.16 is used to obtain strict exterior-angle comparisons. The relation

$$AH \cong AC$$

makes triangle AHC isosceles, allowing the relevant angles at H and C to be transported across congruence.

The point Y lies between B and C, so the appropriate ray from H meets the opposite side internally. The theorem `hilbert_interior_angle_less` turns this incidence fact into a strict angle comparison. After the necessary same-ray normalizations and transitivity steps, the proof obtains

$$\angle BCH < \angle BHC.$$

Proposition I.19 applied to triangle BHC now gives

$$BH < BC.$$

6.4 Case 2: $H = Y$

If the auxiliary point is itself the intersection point with BC, then

$$B - H - C.$$

The definition of strict segment comparison immediately gives

$$BH < BC.$$

No angle comparison is required.

6.5 Case 3: $A - Y - H$

This branch has a different geometry. At C, the ray through Y meets the open segment HA, so

$$\angle HCY < \angle HCA.$$

Because $C - Y - B$, this comparison can be transported from the ray CY to the ray CB.

Again $AH \cong AC$ makes triangle AHC isosceles. The base-angle congruence transfers the comparison to the corresponding angle at H. At H, the ray through Y meets the open segment CB, giving a second proper interior-subangle comparison. Since $H - Y - A$, the ray HY agrees with the ray HA. Transitivity and angle-congruence transport therefore yield

$$\angle BCH < \angle BHC.$$

A final application of I.19 gives

$$BH < BC.$$

The two nontrivial cases therefore reach the same final comparison through different angle-order paths.

6.6 The Hinge Helper

The theorem `euclid_proposition_24_hinge_aux` packages the complete case analysis:

$$\begin{array}{ccc} & \text{interior ray through } H \text{ meets } BC \text{ at } Y & \\ & \downarrow & \\ & \text{same-ray trichotomy for } A, Y, H & \\ & \downarrow & \\ A - H - Y & H = Y & A - Y - H \\ \downarrow & \downarrow & \downarrow \\ \text{Case 1} & \text{betweenness} & \text{Case 3} \\ & \downarrow & \\ & BH < BC. & \end{array}$$

No location of H is inferred from the diagram.

6.7 Completion of I.24

The SAS helper supplies

$$BH \cong EF,$$

and the hinge helper supplies

$$BH < BC.$$

Strict segment comparison is transported through congruence, giving

$$EF < BC.$$

This is exactly the conclusion of Proposition I.24.

7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_24
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hDEF : Not (PrimCollinear Geo D E F))
    (hAB_DE : Geo.Congruent A B D E)
    (hAC_DF : Geo.Congruent A C D F)
    (hAngle :
      HilbertAngleLess Geo E D F B A C) :
    HilbertSegmentLess Geo E F B C := by
  ...
```

The first major helper packages the SAS construction. Its output contains the auxiliary ray and point together with the two metric facts needed later:

$$AH \cong AC, \qquad BH \cong EF.$$

Schematically, the public proof begins by unpacking this construction:

```
rcases
    euclid_proposition_24_sas_aux
      (Geo := Geo)
      A B C D E F
      hABC hDEF hAB_DE hAC_DF hAngle with
  <X, H, hMeet, hRayAXH, hAH_AC, hBH_EF>
```

The hinge comparison is then delegated to the second major helper:

```
have hBH_BC :
    HilbertSegmentLess Geo B H B C :=
  euclid_proposition_24_hinge_aux
    (Geo := Geo)
    A B C H X
    hABC
    hMeet
    hRayAXH
    hAH_AC
```

Inside that helper the intersection witness is extracted and the point order is split exhaustively:

```
rcases hMeet with <Y, hBYC, hRayAXY>

rcases
    euclid_proposition_22_sameRay_trichotomy
      (Geo := Geo)
      A Y H hAY hAH hRayAYH with
  hAHY | hYH | hAYH
```

The three branches are then dispatched to Case 1, the immediate betweenness case, and Case 3. The public theorem contains none of this order analysis; it sees only the reusable result $BH < BC$.

Finally the segment inequality is transported from BH to the congruent segment EF. Thus the public proof is short because the difficult geometry has been isolated in named synthetic helpers.

8 Relationship with Earlier Propositions

8.1 Proposition I.5

The isosceles base-angle theorem is used repeatedly after

$$AH \cong AC.$$

It is the main congruence bridge between the angle at C and the angle at H in the hinge triangle.

8.2 Proposition I.16

Case 1 uses the exterior-angle theorem to obtain strict angle comparisons which are not available directly from the interior-ray configuration. This is the principal reason that Case 1 is longer than Case 3.

8.3 Proposition I.19

Both nontrivial branches end with the same angle inequality in triangle BHC:

$$\angle BCH < \angle BHC.$$

Proposition I.19 converts this angle comparison into the required side comparison

$$BH < BC.$$

Thus I.19 is the final metric step of the hinge argument.

8.4 Proposition I.23

The classical Euclidean proof uses I.23 to construct at D an angle equal to the given larger angle. In the Hilbert reconstruction the corresponding construction is expressed through the already available Hilbert angle construction machinery. Proposition I.23 is therefore part of the classical dependency picture, while the formal proof uses the more general interface theorem behind that construction.

9 Hilbert and Book Zero Components Used

9.1 HilbertAngleLess

Strict angle comparison is the input order relation of I.24. It also provides the interior-ray witness needed for the auxiliary construction.

9.2 HilbertSegmentLess

Strict segment comparison is the conclusion of I.24. Its witness-based form permits the middle case $H = Y$ to close directly from betweenness.

9.3 HilbertSameRay

Same-ray machinery is used both to represent the auxiliary interior ray and to normalize angles when two syntactically different rays determine the same geometric direction.

9.4 Same-ray trichotomy

The trichotomy is the formal mechanism that exposes the hidden case split. It produces the exhaustive alternatives

$$A - H - Y, \qquad H = Y, \qquad A - Y - H.$$

This is the order-theoretic core of the reconstruction.

9.5 hilbert_interior_angle_less

This theorem turns a ray meeting the opposite segment internally into a strict comparison between the resulting subangle and the whole angle. It is used repeatedly in the two nontrivial hinge cases.

9.6 Angle-congruence transport

The formal proof must explicitly transport strict inequalities across congruent angles and across coincident rays. These steps replace informal phrases such as "this is the same angle" which are normally read directly from a diagram.

10 Historical Note: Euclid, Proclus, Simson, and Hilbert

Proposition I.24 is a useful example of the distinction between a classical geometric argument and a complete axiomatic treatment of its diagram.

The traditional Euclidean proof presents one configuration. As emphasized in modern study notes by David R. Wilkins, three configurations can arise, and Euclid's printed argument covers only one of them. Proclus supplied proofs for the two omitted configurations, and later commentators reproduced or reorganized these supplements.

Robert Simson pursued a different repair. He added a condition governing which of the two equal sides is to be used in the construction, thereby seeking to force the auxiliary point into the configuration required by the standard proof. Heath records this modification in his commentary on I.24.

Hilbert's significance is different. His axiomatic treatment of order and incidence supplies a framework in which such positional assumptions cannot remain implicit. Betweenness, collinearity, ray identity, and separation must be justified formally. The Lean reconstruction inherits precisely this discipline.

Accordingly, the three-way split in the present proof should not be read as a peculiarity of proof-assistant programming. It is a machine-checked form of a genuine classical issue in Proposition I.24.

11 Dependency Summary

The formal dependency structure is

$$\begin{gathered}\text{Hilbert incidence + order + congruence}\\ \downarrow\\ \texttt{HilbertAngleLess},\quad \texttt{HilbertSegmentLess},\quad \texttt{HilbertSameRay}\\ \downarrow\\ \text{angle construction, interior-ray theory, same-ray trichotomy}\\ \downarrow\\ \text{I.5, I.16, I.19 and Book Zero transport lemmas}\\ \downarrow\\ \texttt{euclid\_proposition\_24\_sas\_aux}\\ \texttt{euclid\_proposition\_24\_hinge\_case1}\\ \texttt{euclid\_proposition\_24\_hinge\_case3}\\ \texttt{euclid\_proposition\_24\_hinge\_aux}\\ \downarrow\\ \texttt{euclid\_proposition\_24}.\end{gathered}$$

No numerical metric, numerical angle measure, circle continuity, or additional hinge axiom is introduced.

12 Conclusion

Proposition I.24 is one of the clearest examples so far of what formal reconstruction can reveal about a classical synthetic proof.

The metric core is simple. The auxiliary SAS construction gives

$$BH \cong EF,$$

and the final application of Proposition I.19 converts an angle comparison into

$$BH < BC.$$

The real difficulty is positional: the auxiliary point does not have a single order relation forced by the hypotheses.

The Lean proof therefore separates the possibilities

$$A - H - Y, \qquad H = Y, \qquad A - Y - H$$

and proves each one. This exhaustive split corresponds to a difficulty already recognized in the classical commentary tradition. The formal proof does not replace Euclid's synthetic geometry; it supplies the order and incidence information that the traditional diagram leaves implicit.

The completed development compiles cleanly and contains no `sorry`. Proposition I.24 introduces no new geometric axiom. It uses the Hilbert framework, the Book Zero proof language, and the synthetic results established earlier in Book I.